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Organic compound and organic electroluminescent device comprising same

Abstract

A novel compound represented by Chemical Formula 6 below and an organic electroluminescent device including the same are disclosed. The inclusion of the compound in an organic material layer of an organic electroluminescent device, preferably as a light emitting layer, can improve luminous efficiency, driving voltage, lifetime and the like of the organic electroluminescent device:
##STR00001##

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/KR2018/005079 filed May 2, 2018, claiming priority based on Korean Patent Application No. 10-2017-0083287 filed Jun. 30, 2017.

TECHNICAL FIELD

(2) The present invention relates to a novel organic compound capable of being used as a material for an organic electroluminescent device, and an organic electroluminescent device including the same.

BACKGROUND ART

(3) With the observation of organic thin film light emission made by Bemanose in 1950s as a start, studies on organic electroluminescent (EL) devices have been continued leading to blue electroluminescence using a single anthracene crystal in 1965, and in 1987, an organic electroluminescent device having a laminated structure divided into functional layers of a hole layer and a light emitting layer has been proposed by Tang. After that, in order to manufacture organic electroluminescent devices with high efficiency and long lifetime, development has been made in the form of introducing each characteristic organic material layer into the device, which leads to the development of specialized materials used therein.

(4) When a voltage is applied between the two electrodes in an organic electroluminescent device, holes and electrons are injected to an organic material layer from the anode and the cathode, respectively. When the injected holes and electrons meet, excitons are formed, and light emits when these excitons fall back to the ground state. Herein, materials used as the organic material layer may be divided into a light emitting material, a hole injection material, a hole transport material, an electron transport material, an electron injection material and the like depending on the function.

(5) The light emitting material may be divided into, depending on the light emitting color, blue, green and red light emitting materials, and yellow and orange light emitting materials for obtaining better natural colors. In addition, in order to increase color purity and increase luminous efficiency through energy transfer, host/dopant series may be used as the light emitting material.

(6) The dopant material may be divided into fluorescent dopants using organic materials and phosphorescent dopants using metal complex compounds including heavy atoms such as Ir or Pt. Herein, development of phosphorescent materials may enhance luminous efficiency up to 4 times compared to fluorescence theoretically, and therefore, studies on phosphorescent host materials have been widely progressed as well as on phosphorescent dopants.

(7) So far, NPB, BCP, Alq.sub.3 and the like have been widely known as materials of a hole injection layer, a hole transport layer, a hole blocking layer and an electron transport layer, and anthracene derivatives have been reported as a material of a light emitting layer. Particularly, among light emitting layer materials, metal complex compounds including Ir such as Firpic, Ir(ppy).sub.3 or (acac)Ir(btp).sub.2 having advantages in terms of efficiency enhancement have been used as blue, green and red phosphorescent dopant materials, and 4,4-dicarbazolylbiphenyl (CBP) has been used as a phosphorescent host material.

(8) ##STR00002##

(9) However, although being advantageous in terms of a light emission property, existing organic material layer materials have a low glass transition temperature and thereby have very unfavorable thermal stability, which is not satisfactory in terms of an organic electroluminescent device lifetime. Accordingly, development of organic material layer materials having superior performance has been required.

DISCLOSURE

Technical Problem

(10) The present invention is directed to providing a novel organic compound capable of being used in an organic electroluminescent device, and having excellent hole and electron injection and transport abilities, a light emitting ability and the like.

(11) The present invention is also directed to providing an organic electroluminescent device including the novel organic compound, and thereby exhibiting a low driving voltage, high luminous efficiency, and an enhanced lifetime.

Technical Solution

(12) In view of the above, one embodiment of the present invention provides a compound represented by the following Chemical Formula 1:

(13) ##STR00003## in Chemical Formula 1, X.sub.1 is selected from the group consisting of S, O, N(R.sub.1) and C(R.sub.2)(R.sub.3); X.sub.2 and X.sub.3 are each independently N or C(Ar.sub.1), and at least one thereof is C(Ar.sub.1); and ring A is represented by any one of the following Chemical Formulae 2 to 4;

(14) ##STR00004## in Chemical Formulae 1 to 4, a dotted line means a part that is fused; m is an integer of 0 to 4; n is an integer of 0 to 6; R.sub.1 to R.sub.4 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a

C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group; the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.1 to R.sub.4 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other; and Ar.sub.1 is a substituent represented by the following Chemical Formula 5, and when Ar.sub.1 is present in plural numbers, these are the same as or different from each other;

(15) ##STR00005## in Chemical Formula 5, means a part where a bond is formed; L.sub.1 and L.sub.2 are each independently selected from the group consisting of a direct bond, a C.sub.6~C.sub.18 arylene group and a heteroarylene group having 5 to 18 nuclear atoms; Ar.sub.2 is selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group; and the arylene group and the heteroarylene group of L.sub.1 and L.sub.2, and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of Ar.sub.2 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(16) Another embodiment of the present invention provides an organic electroluminescent device including an anode, a cathode, and one or more organic material layers provided between the anode and the cathode, wherein at least one of the one or more organic material layers includes the compound of Chemical Formula 1.

(17) The “alkyl” in the present invention is a monovalent substituent derived from linear or branched saturated hydrocarbon having 1 to 40 carbon atoms. Examples thereof may include

methyl, ethyl, propyl, isobutyl, sec-butyl, pentyl, iso-amyl, hexyl and the like, but are not limited thereto.

(18) The “alkenyl” in the present invention is a monovalent substituent derived from linear or branched unsaturated hydrocarbon having one or more carbon-carbon double bonds and having 2 to 40 carbon atoms. Examples thereof may include vinyl, allyl, isopropenyl, 2-butenyl and the like, but are not limited thereto.

(19) The “alkynyl” in the present invention is a monovalent substituent derived from linear or branched unsaturated hydrocarbon having one or more carbon-carbon triple bonds and having 2 to 40 carbon atoms. Examples thereof may include ethynyl, 2-propynyl and the like, but are not limited thereto.

(20) The “aryl” in the present invention means a monovalent substituent derived from aromatic hydrocarbon having a single ring or two or more rings combined and having 6 to 60 carbon atoms. In addition, a monovalent substituent having two or more rings fused with each other, including only carbon (for example, the number of carbon atoms may be from 8 to 60) as a ring-forming atom, and with the whole molecule having non-aromaticity may also be included. Examples of such aryl may include phenyl, naphthyl, phenanthryl, anthryl, fluorenyl and the like, but are not limited thereto. The “heteroaryl” in the present invention means a monovalent substituent derived from monoheterocyclic or polyheterocyclic aromatic hydrocarbon having 5 to 60 nuclear atoms. Herein, one or more carbons, preferably 1 to 3 carbons, in the ring are substituted with a heteroatom selected from among N, O, P, S and Se. In addition, a monovalent group having two or more rings simply attached (pendant) or fused with each other, including a heteroatom selected from among N, O, P, S and Se as a ring-forming atom in addition to carbon, and with the whole molecule having non-aromaticity is interpreted to be included as well. Examples of such heteroaryl may include 6-membered monocyclic rings such as pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl or triazinyl; polycyclic rings such as phenoxathienyl, indoliziny, indolyl, purinyl, quinolyl, benzothiazole or carbazolyl; 2-furanyl, N-imidazolyl, 2-isoxazolyl, 2-pyridinyl, 2-pyrimidinyl and the like, but are not limited thereto.

(21) The “aryloxy” in the present invention is a monovalent substituent represented by RO—, and R means aryl having 5 to 60 carbon atoms. Examples of such aryloxy may include phenyloxy, naphthyloxy, diphenyloxy and the like, but are not limited thereto.

(22) The “alkyloxy” in the present invention is a monovalent substituent represented by R'O—, and R' means alkyl having 1 to 40 carbon atoms and is interpreted to include a linear, branched or cyclic structure. Examples of such alkyloxy may include methoxy, ethoxy, n-propoxy, 1-propoxy, t-butoxy, n-butoxy, pentoxy and the like, but are not limited thereto.

(23) The “arylamine” in the present invention means amine substituted with aryl having 6 to 60 carbon atoms.

(24) The “cycloalkyl” in the present invention means a monovalent substituent derived from monocyclic or polycyclic non-aromatic hydrocarbon having 3 to 40 carbon atoms. Examples of such cycloalkyl may include cyclopropyl, cyclopentyl, cyclohexyl, norbomyl, adamantane and the like, but are not limited thereto.

(25) The “heterocycloalkyl” in the present invention means a monovalent substituent derived from non-aromatic hydrocarbon having 3 to 40 nuclear atoms, and one or more carbons, preferably 1 to 3 carbons, in the ring are substituted with a heteroatom such as N, O, S or Se. Examples of such heterocycloalkyl may include morpholine, piperazine and the like, but are not limited thereto.

(26) The “alkylsilyl” in the present invention means silyl substituted with alkyl having 1 to 40 carbon atoms, and the “arylsilyl” means silyl substituted with aryl having 5 to 60 carbon atoms.

(27) The “fused ring” in the present invention means a fused aliphatic ring, a fused aromatic ring, a fused heteroaliphatic ring, a fused heteroaromatic ring, or a combined form thereof.

Advantageous Effects

(28) A compound of the present invention has excellent thermal stability, carrier transport ability,

light emitting ability and the like, and therefore, is useful as a material of an organic material layer of an organic electroluminescent device.

(29) In addition, an organic electroluminescent device including a compound of the present invention in an organic material layer exhibits greatly enhanced properties in terms of light emitting performance, driving voltage, lifetime, efficiency and the like, and can be effectively used in a full color display panel and the like.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a sectional diagram illustrating an organic electroluminescent device according to one embodiment of the present invention.

(2) FIG. 2 is a sectional diagram illustrating an organic electroluminescent device according to one embodiment of the present invention.

MODE FOR DISCLOSURE

(3) Hereinafter, the present invention will be described in detail.

(4) 1. Novel Organic Compound

(5) A novel compound of the present invention may be represented by the following Chemical Formula 1:

(6) ##STR00006## in Chemical Formula 1, X.sub.1 is selected from the group consisting of S, O, N(R.sub.1) and C(R.sub.2)(R.sub.3); X.sub.2 and X.sub.3 are each independently N or C(Ar.sub.1), and at least one thereof is C(Ar.sub.1); and ring A is represented by any one of the following Chemical Formulae 2 to 4;

(7) ##STR00007## in Chemical Formulae 1 to 4, a dotted line means a part that is fused; m is an integer of 0 to 4; n is an integer of 0 to 6; R.sub.1 to R.sub.4 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group; the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.1 to R.sub.4 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other; and Ar.sub.1 is a substituent represented by the following Chemical Formula 5, and when Ar.sub.1 is present in plural numbers, these are the same as or different from each other;

(8) ##STR00008## in Chemical Formula 5, * means a part where a bond is formed; L.sub.1 and

L.sub.2 are each independently selected from the group consisting of a direct bond, a C.sub.6~C.sub.18 arylene group and a heteroarylene group having 5 to 18 nuclear atoms; Ar.sub.2 is selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group; and the arylene group and the heteroarylene group of L.sub.1 and L.sub.2, and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of Ar.sub.2 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(9) More specifically, the compound represented by Chemical Formula 1 of the present invention has an EWG bonding to a 5-membered aromatic ring or a 5-membered aromatic heteroring such as indole, indazole, indene, benzofuran, benzothiophene or triazolo, and may be used as a host material since, by having a similar energy level with carbazole, the energy level may be adjusted to be higher than a dopant energy level. Particularly, benzofuran and benzothiophene moieties are electron abundant, and when used as an electron transport layer material of an organic electroluminescent device, mobility increases, and as a result, an increase in the luminous efficiency and a decrease in the driving voltage may be expected. In addition, the 5-membered aromatic ring or the 5-membered aromatic heteroring of the present invention has a smaller molecular weight compared to existing compounds, and therefore, may be deposited at a relatively lower deposition temperature when deposited compared to other materials, and favorable processability and enhanced thermal stability may be obtained.

(10) Accordingly, the compound represented by Chemical Formula 1 of the present invention may be used as a material of an organic material layer of an organic electroluminescent device, and may be preferably used as a light emitting layer material (green phosphorescent host material), an electron transport layer/injection layer material, a light emitting auxiliary layer material or an electron transport auxiliary layer material, and more preferably as a light emitting layer material, an electron transport layer material or an electron transport auxiliary layer material. In addition, the organic electroluminescent device including the compound of Chemical Formula 1 may have performance and lifetime properties greatly enhanced, and a full color organic light emitting panel using such an organic electroluminescent device may have its performance maximized.

(11) According to preferred one embodiment of the present invention, the compound may be represented by any one of the following Chemical Formulae 6 to 14:

(12) ##STR00009## in Chemical Formulae 6 to 14, ring A, Ar.sub.1 and R.sub.1 to R.sub.3 each have the same definitions as in Chemical Formula 1.

(13) According to preferred one embodiment of the present invention, the compound is preferably represented by any one of Chemical Formulae 6 to 8 in terms of securing low driving voltage and high luminous efficiency, and may be more preferably represented by Chemical Formula 6 or 7, and even more preferably represented by Chemical Formula 8 or 10.

(14) According to preferred one embodiment of the present invention, R.sub.1 to R.sub.4 are each independently selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and the alkyl group, the aryl group and the heteroaryl group of R.sub.1 to R.sub.4 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these may be the same as or different from each other.

(15) According to preferred one embodiment of the present invention, R.sub.1 to R.sub.4 are each independently selected from the group consisting of a methyl group, an ethyl group, a propyl group, a butyl group, a pentyl group, a phenyl group, a biphenyl group, a terphenyl group, a naphthalenyl group, a pyridinyl group, a pyrimidinyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a carbazolyl group, a fluorenyl group, a spirofluorenyl group and a dibenzodioxynyl group, and the methyl group, the ethyl group, the propyl group, the butyl group, the pentyl group, the phenyl group, the biphenyl group, the terphenyl group, the naphthalenyl group, the pyridinyl group, the pyrimidinyl group, the triazinyl group, the dibenzofuranyl group, the dibenzothiophenyl group, the carbazolyl group, the fluorenyl group, the spirofluorenyl group and the dibenzodioxynyl group of R.sub.1 to R.sub.4 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 arylamine group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these may be the same as or different from each other.

(16) According to preferred one embodiment of the present invention, L.sub.1 and L.sub.2 are each independently a direct bond, or a linker represented by any one of the following Chemical Formulae A-1 to A-7, but are not limited thereto:

(17) ##STR00010## in Chemical Formulae A-1 to A-7, * means a part where a bond is formed; Z.sub.1 to Z.sub.8 are each independently N or C(R.sub.5); any two of Z.sub.1 to Z.sub.6 forming a bond as a linker in Chemical Formula A-1 are C(R.sub.5), and herein, R.sub.5 is not present; any one of Z.sub.1 to Z.sub.4 and any one of Z.sub.5 to Z.sub.8 forming a bond as a linker in Chemical Formula A-4 are C(R.sub.5), and herein, R.sub.5 is not present; any two of Z.sub.1 to Z.sub.4 forming a bond as a linker in Chemical Formula A-5 are C(R.sub.5), and herein, R.sub.5 is not present; X.sub.4 and X.sub.5 are each independently O, S, N(R.sub.6) or C(R.sub.7)(R.sub.8); X.sub.6 is N or C(R.sub.9); R.sub.5 to R.sub.9 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group, or adjacent groups bond to form a fused ring, and when R.sub.5 to R.sub.8 are each present in plural numbers, these are the same as or different from each other; and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.5 to R.sub.9 are each

independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(18) According to preferred one embodiment of the present invention, L.sub.1 and L.sub.2 are each independently a direct bond, or a linker represented by any one of the following Chemical Formulae B-1 to B-12, but are not limited thereto:

(19) ##STR00011## in Chemical Formulae B-1 to B-12, * means a part where a bond is formed; R.sub.6 to R.sub.9 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group; and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.6 to R.sub.9 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(20) According to preferred one embodiment of the present invention, in Chemical Formulae B-1 to B-12, R.sub.6 to R.sub.9 are each independently selected from the group consisting of hydrogen, a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms; and the alkyl group, the aryl group and the heteroaryl group of R.sub.6 to R.sub.9 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(21) According to preferred one embodiment of the present invention, L.sub.1 and L.sub.2 are each independently a direct bond, or may be a linker selected from the group consisting of Chemical Formulae B-1 to B-3 and B-7 to B-12.

(22) According to preferred one embodiment of the present invention, Ar.sub.2 is a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, or an mylamine

group, and the aryl group, the heteroaryl group and the mylamine group of Ar.sub.2 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 mylamine group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these may be the same as or different from each other.

(23) According to preferred one embodiment of the present invention, Ar.sub.2 may be a substituent represented by the following Chemical Formula 15:

(24) ##STR00012## in Chemical Formula 15, * means a part where a bond is formed; Y.sub.1 to Y.sub.5 are each independently N or C(R.sub.10); R.sub.10 is selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group, and when R.sub.10 is present in plural numbers, these are the same as or different from each other; and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.10 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(25) According to preferred one embodiment of the present invention, the substituent represented by Chemical Formula 15 may be a substituent represented by the following Chemical Formula 16:

(26) ##STR00013## in Chemical Formula 16, * means a part where a bond is formed; R.sub.11 and R.sub.12 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group, or bond to adjacent groups to form a fused ring; the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.11 and R.sub.12 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a

C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other; and Y.sub.1, Y.sub.3 and Y.sub.5 each have the same definitions as in Chemical Formula 15. (27) According to preferred one embodiment of the present invention, the substituent represented by Chemical Formula 15 may be a substituent represented by any one of the following Chemical Formulae C-1 to C-5:

(28) ##STR00014## in Chemical Formulae C-1 to C-5, * means a part where a bond is formed, R.sub.11 and R.sub.12 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group, or bond to adjacent groups to form a fused ring; and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.11 and R.sub.12 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(29) According to preferred one embodiment of the present invention, R.sub.11 and R.sub.12 are each independently selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and the alkyl group, the aryl group and the heteroaryl group of R.sub.11 and R.sub.12 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(30) According to preferred one embodiment of the present invention, R.sub.11 and R.sub.12 are each independently selected from the group consisting of a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothiophenyl group of R.sub.11 and R.sub.12 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a

C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(31) According to preferred one embodiment of the present invention, R.sub.11 and R.sub.12 are each independently selected from the group consisting of a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothiophenyl group of R.sub.11 and R.sub.12 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a methyl group, an ethyl group, a propanyl group, a butyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(32) According to preferred one embodiment of the present invention, Ar.sub.2 may be a substituent represented by the following Chemical Formula 17:

(33) ##STR00015## in Chemical Formula 17, * means a part where a bond is formed; R.sub.13 and R.sub.14 are each independently selected from the group consisting of hydrogen, deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 aryloxy group, a C.sub.3~C.sub.40 alkylsilyl group, a C.sub.6~C.sub.60 arylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylamine group, or bond to adjacent groups to form a fused ring; and the alkyl group, the alkenyl group, the alkynyl group, the aryl group, the heteroaryl group, the aryloxy group, the alkyloxy group, the cycloalkyl group, the heterocycloalkyl group, the arylamine group, the alkylsilyl group, the alkylboron group, the arylboron group, the arylphosphanyl group, the mono or diarylphosphinyl group and the arylsilyl group of R.sub.13 and R.sub.14 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of deuterium, halogen, a cyano group, a nitro group, a C.sub.1~C.sub.40 alkyl group, a C.sub.2~C.sub.40 alkenyl group, a C.sub.2~C.sub.40 alkynyl group, a C.sub.6~C.sub.60 aryl group, a heteroaryl group having 5 to 60 nuclear atoms, a C.sub.6~C.sub.60 aryloxy group, a C.sub.1~C.sub.40 alkyloxy group, a C.sub.6~C.sub.60 arylamine group, a C.sub.3~C.sub.40 cycloalkyl group, a heterocycloalkyl group having 3 to 40 nuclear atoms, a C.sub.1~C.sub.40 alkylsilyl group, a C.sub.1~C.sub.40 alkylboron group, a C.sub.6~C.sub.60 arylboron group, a C.sub.6~C.sub.60 arylphosphanyl group, a C.sub.6~C.sub.60 mono or diarylphosphinyl group and a C.sub.6~C.sub.60 arylsilyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(34) According to preferred one embodiment of the present invention, R.sub.13 and R.sub.14 are each independently selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and the alkyl group, the aryl group and the heteroaryl group of R.sub.13 and R.sub.14 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(35) According to preferred one embodiment of the present invention, R.sub.13 and R.sub.14 are each independently selected from the group consisting of a phenyl group, a biphenyl group, a

pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothiophenyl group of R.sub.13 and R.sub.14 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a C.sub.1~C.sub.40 alkyl group, a C.sub.6~C.sub.60 aryl group and a heteroaryl group having 5 to 60 nuclear atoms, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(36) According to preferred one embodiment of the present invention, R.sub.13 and R.sub.14 are each independently selected from the group consisting of a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and the phenyl group, the biphenyl group, the pyridinyl group, the pyrimidinyl group, the dibenzofuranyl group, the carbazolyl group, the fluorenyl group and the dibenzothiophenyl group of R.sub.13 and R.sub.14 are each independently unsubstituted or substituted with one or more types of substituents selected from the group consisting of a methyl group, an ethyl group, a propanyl group, a butyl group, a phenyl group, a biphenyl group, a pyridinyl group, a pyrimidinyl group, a dibenzofuranyl group, a carbazolyl group, a fluorenyl group and a dibenzothiophenyl group, and when substituted with a plurality of the substituents, these are the same as or different from each other.

(37) The compound represented by Chemical Formula 1 of the present invention may be represented by the following compounds, but is not limited thereto:

(38) ##STR00016## ##STR00017## ##STR00018## ##STR00019## ##STR00020##
##STR00021## ##STR00022## ##STR00023## ##STR00024## ##STR00025## ##STR00026##
##STR00027## ##STR00028## ##STR00029## ##STR00030## ##STR00031## ##STR00032##
##STR00033## ##STR00034## ##STR00035## ##STR00036## ##STR00037## ##STR00038##
##STR00039## ##STR00040## ##STR00041## ##STR00042## ##STR00043##

(39) The compound of Chemical Formula 1 of the present invention may be synthesized using general synthesis methods (refer to *Chem. Rev.*, 60:313 (1960); *J. Chem. SOC.* 4482 (1955); *Chem. Rev.* 95: 2457 (1995) or the like). Detailed synthesis processes of the compounds of the present invention will be specifically described in synthesis examples to be described later.

(40) 2. Organic Electroluminescent Device

(41) Meanwhile, another aspect of the present invention relates to an organic electroluminescent device (organic EL device) including the compound represented by Chemical Formula 1 according to the present invention.

(42) Specifically, the present invention relates to an organic electroluminescent device including an anode, a cathode, and one or more organic material layers provided between the anode and the cathode, and at least one of the one or more organic material layers includes the compound represented by Chemical Formula 1. Herein, the compound may be used either alone or as a mixture of two or more.

(43) The one or more organic material layers may be any one or more of a hole injection layer, a hole transport layer, a light emitting layer, a light emitting auxiliary layer, a lifetime improving layer, an electron transport layer, an electron transport auxiliary layer and an electron injection layer, and at least one organic material layer among these may include the compound represented by Chemical Formula 1.

(44) The structure of the organic electroluminescent device according to the present invention described above is not particularly limited, but, when referring to FIG. 1 as one example, includes an anode (10) and a cathode (20) facing each other, and an organic layer (30) located between the anode (10) and the cathode (20). Herein, the organic layer (30) may include a hole transport layer (31), a light emitting layer (32) and an electron transport layer (34). In addition, a hole transport auxiliary layer (33) may be included between the hole transport layer (31) and the light emitting

layer (32), and an electron transport auxiliary layer (35) may be included between the electron transport layer (34) and the light emitting layer (32).

(45) When referring to FIG. 2 as another example of the present invention, the organic layer (30) may further include a hole injection layer (37) between the hole transport layer (31) and the anode (10), and may further include an electron injection layer (36) between the electron transport layer (34) and the cathode (20).

(46) The hole injection layer (37) laminated between the hole transport layer (31) and the anode (10) in the present invention is a layer having a function of, as well as improving interfacial properties between ITO used as the anode and an organic material used as the hole transport layer (31), smoothing the ITO surface by being coated on the top of the ITO of which surface is not smooth, and those commonly used in the art may be used without particular limit, and for example, amine compounds may be used. However, the hole injection layer is not limited thereto.

(47) In addition, the electron injection layer (36) is a layer laminated on the top of the electron transport layer and having a function of facilitating electron injection from the cathode and eventually improving power efficiency, and is not particularly limited as long as it is commonly used in the art. For example, materials such as LiF, Liq, NaCl, CsF, Li.sub.2O or BaO may be used.

(48) Although not shown in the drawings in the present invention, a light emitting auxiliary layer may be further included between the hole transport auxiliary layer (33) and the light emitting layer (32). The light emitting auxiliary layer may perform a role of adjusting a thickness of the organic layer (30) while performing a role of transporting holes to the light emitting layer (32). The light emitting auxiliary layer may include a hole transport material, and may be formed with the same material as the hole transport layer (31).

(49) In addition, although not shown in the drawings in the present invention, a lifetime improving layer may be further included between the electron transport auxiliary layer (35) and the light emitting layer (32). Holes migrating to the light emitting layer (32) by getting on an ionization potential level in an organic light emitting device are not able to diffuse or migrate to the electron transport layer by being blocked by a high energy barrier of the lifetime improving layer, and consequently, the lifetime improving layer has a function of limiting the holes in the light emitting layer. Such a function of limiting the holes in the light emitting layer prevents the holes from diffusing to the electron transport layer migrating electrons by reduction, and therefore, suppresses a lifetime decrease phenomenon caused through an irreversible decomposition reaction by oxidation, and thereby contributes to improving a lifetime of the organic light emitting device.

(50) The compound represented by Chemical Formula 1 in the present invention has an EWG bonding to a 5-membered aromatic ring or a 5-membered aromatic heteroring such as indole, indazole, indene, benzofuran, benzothiophene or triazolo, and may be used as a host material since, by having a similar energy level with carbazole, the energy level may be adjusted to be higher than a dopant energy level. Particularly, benzofuran and benzothiophene moieties are electron abundant, and when used as an electron transport layer material of an organic electroluminescent device, mobility increases, and as a result, an increase in the luminous efficiency and a decrease in the driving voltage may be expected. In addition, the 5-membered aromatic ring or the 5-membered aromatic heteroring of the present invention has a smaller molecular weight compared to existing compounds, and therefore, may be deposited at a relatively lower deposition temperature when deposited compared to other materials, and favorable processability and enhanced thermal stability may be obtained.

(51) Accordingly, the compound represented by Chemical Formula 1 of the present invention may be used as a material of any one of a hole injection layer, a hole transport layer, a light emitting layer, an electron transport layer and an electron injection layer, an organic material layer of an organic electroluminescent device, but may be preferably used as a material of any of a light emitting layer, an electron transport layer, and an electron transport auxiliary layer further laminated on the electron transport layer, and more preferably used as a material of an electron

transport layer or an electron transport auxiliary layer.

(52) In addition, when using the compound according to the present invention as a light emitting layer material, the compound represented by Chemical Formula 1 may be specifically used as a phosphorescent host, a fluorescent host or a dopant material of the light emitting layer, and may be preferably used as a phosphorescent host (blue, green and/or red phosphorescent host material).

(53) In addition, the organic electroluminescent device in the present invention has, as described above, an anode, one or more organic material layers and a cathode consecutively laminated, and in addition thereto, may further include an insulating layer or an adhesive layer at an interface between the electrode and the organic material layer.

(54) Except that at least one or more of the organic material layers (for example, electron transport auxiliary layer) are formed to include the compound represented by Chemical Formula 1, the organic electroluminescent device of the present invention may be manufactured by forming other organic material layers and electrodes using materials and methods known in the art.

(55) The organic material layer may be formed using a vacuum deposition method or a solution coating method. Examples of the solution coating method may include spin coating, dip coating, doctor blading, inkjet printing, thermal transfer method or the like, but are not limited thereto.

(56) A substrate capable of being used in the present invention is not particularly limited, and silicon wafers, quartz, glass plates, metal plates, plastic films, sheets and the like may be used.

(57) The anode material may be prepared using, for example, a conductor having high work function so as to have smooth hole injection, and examples thereof may include metals such as vanadium, chromium, copper, zinc or gold, or alloys thereof; metal oxides such as zinc oxide, indium oxide, indium tin oxide (ITO) or indium zinc oxide (IZO); combinations of metals and oxides such as ZnO:Al or SnO₂:Sb; conductive polymers such as polythiophene, poly(3-methylthiophene), poly[3,4-(ethylene-1,2-dioxy)thiophene] (PEDT), polypyrrole or polyaniline; carbon black, and the like, but are not limited thereto.

(58) The cathode material may be prepared using, for example, a conductor having low work function so as to have smooth electron injection, and examples thereof may include metals such as magnesium, calcium, sodium, potassium, titanium, indium, yttrium, lithium, gadolinium, aluminum, silver, tin or lead, or alloys thereof; and multilayer-structured materials such as LiF/Al or LiO₂/Al, but are not limited thereto.

(59) Hereinafter, the present invention will be described in detail with reference to examples as follows. However, the following examples are for illustrative purposes only, and the present invention is not limited to the following examples.

[Preparation Example 1] Synthesis of A1

(60) ##STR00044##

(61) Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-indazole (8.5 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂·2H₂O (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(62) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A1 (6.8 g, 17.1 mmol, yield 70%).

(63) GC-Mass (theoretical value: 396.3 g/mol, measured value: 396 g/mol)

(64) ¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 8H), 7.62 (s, 1H), 7.84~9.2 (m, 4H)

[Preparation Example 2] Synthesis of A2

(65) ##STR00045##

(66) Under a nitrogen stream, 3-(3-bromophenyl)-1-(pyridin-4-yl)-1H-indazole (8.5 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂·2H₂O (0.6 g, 0.7 mmol), KOAc (7.2 g 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(67) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A2 (7.3 g, 18.3 mmol, yield 75%).

(68) GC-Mass (theoretical value: 397.3 g/mol, measured value: 397 g/mol)

(69) ¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 5H), 7.62 (s, 1H), 7.84~9.2 (m, 4H), 8.11~12 (d, 2H)

[Preparation Example 3] Synthesis of A3

(70) ##STR00046##

(71) Under a nitrogen stream, 3-(3-bromophenyl)-1,5-diphenyl-1H-indazole (10.4 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(72) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A3 (8.6 g, 18.3 mmol, yield 75%).

(73) GC-Mass (theoretical value: 472.4 g/mol, measured value: 472 g/mol)

(74) ¹H-NMR: δ 1.25 (s, 12H), 7.44~7.59 (m, 14H), 7.62 (s, 1H), 8.23~25 (d, 2H)

[Preparation Example 4] Synthesis of A4

(75) ##STR00047##

(76) Under a nitrogen stream, 1-(3-bromophenyl)-3-phenyl-3H-benzo[e]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(77) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A4 (7.6 g, 17.1 mmol, yield 70%).

(78) GC-Mass (theoretical value: 446.4 g/mol, measured value: 446 g/mol)

(79) ¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 10H), 7.62 (s, 1H), 7.81~9.9 (m, 4H)

[Preparation Example 5] Synthesis of A5

(80) ##STR00048##

(81) Under a nitrogen stream, 1-(3-bromophenyl)-3-(pyridin-4-yl)-3H-benzo[e]indazole (9.8 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(82) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A5 (7.6 g, 17.1 mmol, yield 70%).

(83) GC-Mass (theoretical value: 447.4 g/mol, measured value: 447 g/mol)

(84) ¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 9H), 7.62 (s, 1H), 7.81~9.9 (m, 2H), 8.21~23 (d, 2H)

[Preparation Example 6] Synthesis of A6

(85) ##STR00049##

(86) Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[f]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(87) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A6 (7.6 g, 17.1 mmol, yield 70%).

(88) GC-Mass (theoretical value: 446.4 g/mol, measured value: 446 g/mol)

(89) ¹H-NMR: δ 1.25 (s, 12H), 7.45~7.55 (m, 10H), 7.58 (s, 1H), 7.61 (s, 1H), 7.78 (s, 1H), 7.81~8.6 (m, 2H)

[Preparation Example 7] Synthesis of A7

(90) ##STR00050##

(91) Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[g]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(92) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO.sub.4, and the result was purified using column chromatography to obtain target Compound A7 (7.6 g, 17.1 mmol, yield 70%).

(93) GC-Mass (theoretical value: 446.36 g/mol, measured value: 446 g/mol)

(94) 1H-NMR: δ 1.25 (s, 12H), 7.45~7.54 (m, 10H), 7.62 (s, 1H), 7.81~88 (m, 4H)

[Preparation Example 8] Synthesis of A8

(95) ##STR00051##

(96) Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-indole (8.5 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(97) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO.sub.4, and the result was purified using column chromatography to obtain target Compound A8 (6.7 g, 17.1 mmol, yield 70%).

(98) GC-Mass (theoretical value: 395.3 g/mol, measured value: 395 g/mol)

(99) 1H-NMR: δ 1.25 (s, 12H), 7.43~7.55 (m, 9H), 7.62 (s, 1H), 7.84~92 (m, 4H)

[Preparation Example 9] Synthesis of A9

(100) ##STR00052##

(101) Under a nitrogen stream, 3-(3-bromophenyl)-1-(pyridin-4-yl)-1H-indole (8.5 g 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(102) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO.sub.4, and the result was purified using column chromatography to obtain target Compound A9 (6.8 g, 17.1 mmol, yield 70%).

(103) GC-Mass (theoretical value: 396.3 g/mol, measured value: 396 g/mol)

(104) 1H-NMR: δ 1.25 (s, 12H), 7.41~7.55 (m, 6H), 7.62 (s, 1H), 7.84~92 (m, 4H), 8.11~12 (d, 2H)

[Preparation Example 10] Synthesis of A10

(105) ##STR00053##

(106) Under a nitrogen stream, 3-(3-bromophenyl)-1,5-diphenyl-1H-indole (10.3 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(107) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO.sub.4, and the result was purified using column chromatography to obtain target Compound A10 (8.6 g, 18.3 mmol, yield 75%).

(108) GC-Mass (theoretical value: 471.4 g/mol, measured value: 471 g/mol)

(109) 1H-NMR: δ 1.25 (s, 12H), 7.41~7.60 (m, 15H), 7.63 (s, 1H), 8.23~25 (d, 2H)

[Preparation Example 11] Synthesis of A11

(110) ##STR00054##

(111) Under a nitrogen stream, 1-(3-bromophenyl)-3-phenyl-3H-benzo[e]indole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(112) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A11 (7.6 g, 17.1 mmol, yield 70%).

(113) GC-Mass (theoretical value: 445.4 g/mol, measured value: 445 g/mol)

(114) ¹H-NMR: δ 1.25 (s, 12H), 7.41~7.55 (m, 11H), 7.62 (s, 1H), 7.81~7.99 (m, 4H)

[Preparation Example 12] Synthesis of A12

(115) ##STR00055##

(116) Under a nitrogen stream, 1-(3-bromophenyl)-3-(pyridin-4-yl)-3H-benzo[e]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(117) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A12 (7.6 g, 17.1 mmol, yield 70%).

(118) GC-Mass (theoretical value: 446.4 g/mol, measured value: 446 g/mol)

(119) ¹H-NMR: δ 1.25 (s, 12H), 7.42~7.55 (m, 10H), 7.62 (s, 1H), 7.81~7.99 (m, 2H), 8.21~8.23 (d, 2H)

[Preparation Example 13] Synthesis of A13

(120) ##STR00056##

(121) Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[f]indole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(122) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A13 (7.6 g, 17.1 mmol, yield 70%).

(123) GC-Mass (theoretical value: 445.4 g/mol, measured value: 445 g/mol)

(124) ¹H-NMR: δ 1.25 (s, 12H), 7.42~7.55 (m, 11H), 7.58 (s, 1H), 7.61 (s, 1H), 7.78 (s, 1H), 7.81~7.85 (m, 2H)

[Preparation Example 14] Synthesis of A14

(125) ##STR00057##

(126) Under a nitrogen stream, 3-(3-bromophenyl)-1-phenyl-1H-benzo[g]indazole (9.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(127) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A14 (7.6 g, 17.1 mmol, yield 70%).

(128) GC-Mass (theoretical value: 445.37 g/mol, measured value: 445 g/mol)

(129) ¹H-NMR: δ 1.25 (s, 12H), 7.42~7.54 (m, 11H), 7.62 (s, 1H), 7.82~7.89 (m, 4H)

[Preparation Example 15] Synthesis of A15

(130) ##STR00058##

(131) Under a nitrogen stream, 2-(3-bromophenyl)benzofuran (6.7 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl₂.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(132) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO₄, and the result was purified using column chromatography to obtain target Compound A15 (5.5 g, 17.1 mmol, yield 70%).

(133) GC-Mass (theoretical value: 320.2 g/mol, measured value: 320 g/mol)

(134) ¹H NMR: δ 1.25 (s, 12H), 7.11 (s, 1H), 7.58 (s, 1H), 7.67~7.89 (m, 7H)

[Preparation Example 16] Synthesis of A16

(135) ##STR00059##

(136) Under a nitrogen stream, 2-(3-bromophenyl)-1,1-dimethyl-1H-indene (7.3 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl.sub.2 0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(137) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO.sub.4, and the result was purified using column chromatography to obtain target Compound A16 (5.9 g, 17.1 mmol, yield 70%).

(138) GC-Mass (theoretical value: 346.28 g/mol, measured value: 346 g/mol)

(139) ¹H-NMR: δ 1.25 (s, 12H), 1.69 (s, 6H), 7.01 (s, 1H), 7.45 (s, 1H), 7.57~7.77 (m, 7H)

[Preparation Example 17] Synthesis of A17

(140) ##STR00060##

(141) Under a nitrogen stream, 2-(3-bromophenyl)benzo[b]thiophene (7.0 g, 24.4 mmol), 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (7.4 g, 29.2 mmol), Pd(dppf)Cl.sub.2 (0.6 g, 0.7 mmol), KOAc (7.2 g, 73.1 mmol) and 1,4-dioxane (200 ml) were mixed, and stirred for 6 hours at 130° C.

(142) After the reaction was finished, the result was extracted with ethyl acetate, the moisture was removed using MgSO.sub.4, and the result was purified using column chromatography to obtain target Compound A17 (6.1 g, 18.3 mmol, yield 75%).

(143) GC-Mass (theoretical value: 336.26 g/mol, measured value: 336 g/mol)

(144) ¹H-NMR: δ 1.25 (s, 12H), 7.51 (s, 1H), 7.59 (s, 1H), 7.72~7.92 (m, 7H)

[Synthesis Example 1] Synthesis of R1

(145) ##STR00061##

(146) Under a nitrogen stream, A1 (6.8 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(147) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R1 (8.4 g, 12.8 mmol, yield 75%).

(148) GC-Mass (theoretical value: 653.8 g/mol, measured value: 653 g/mol)

[Synthesis Example 2] Synthesis of R8

(149) ##STR00062##

(150) Under a nitrogen stream, A1 (6.8 g 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(151) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R8 (9.0 g, 13.1 mmol, yield 77%).

(152) GC-Mass (theoretical value: 681.8 g/mol, measured value: 682 g/mol)

[Synthesis Example 3] Synthesis of R21

(153) ##STR00063##

(154) Under a nitrogen stream, A2 (6.8 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3

hours at 110° C.

(155) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R21 (7.8 g, 11.9 mmol, yield 70%).

(156) GC-Mass (theoretical value: 654.8 g/mol, measured value: 655 g/mol)

[Synthesis Example 4] Synthesis of R41

(157) ##STR00064##

(158) Under a nitrogen stream, A3 (8.6 g, 18.3 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (9.3 g, 20.1 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 73.1 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(159) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R41 (10.0 g, 13.7 mmol, yield 75%).

(160) GC-Mass (theoretical value: 729.9 g/mol, measured value: 730 g/mol)

[Synthesis Example 5] Synthesis of R61

(161) ##STR00065##

(162) Under a nitrogen stream, A4 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(163) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R61 (9.0 g, 12.8 mmol, yield 75%).

(164) GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 6] Synthesis of R68

(165) ##STR00066##

(166) Under a nitrogen stream, A4 (7.6 g, 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(167) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R68 (9.4 g, 12.8 mmol, yield 75%).

(168) GC-Mass (theoretical value: 731.8 g/mol, measured value: 731 g/mol)

[Synthesis Example 7] Synthesis of R81

(169) ##STR00067##

(170) Under a nitrogen stream, A5 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(171) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R81 (8.9 g 12.6 mmol, yield 74%).

(172) GC-Mass (theoretical value: 703.9 g/mol, measured value: 704 g/mol)

[Synthesis Example 8] Synthesis of R101

(173) ##STR00068##

(174) Under a nitrogen stream, A6 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(175) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R101 (9.0 g, 12.8 mmol, yield 75%).

(176) GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 9] Synthesis of R121

(177) ##STR00069##

(178) Under a nitrogen stream, A7 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(179) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R121 (8.9 g, 12.0 mmol, yield 74%).

(180) GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 10] Synthesis of R141

(181) ##STR00070##

(182) Under a nitrogen stream, A8 (6.7 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(183) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R141 (8.3 g, 12.8 mmol, yield 75%).

(184) GC-Mass (theoretical value: 652.8 g/mol, measured value: 653 g/mol)

[Synthesis Example 11] Synthesis of R148

(185) ##STR00071##

(186) Under a nitrogen stream, A8 (6.7 g, 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(187) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R148 (8.7 g, 12.8 mmol, yield 75%).

(188) GC-Mass (theoretical value: 680.8 g/mol, measured value: 681 g/mol)

[Synthesis Example 12] Synthesis of R161

(189) ##STR00072##

(190) Under a nitrogen stream, A9 (6.8 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(191) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R161 (7.9 g, 11.1 mmol, yield 71%).

(192) GC-Mass (theoretical value: 653.8 g/mol, measured value: 654 g/mol)

[Synthesis Example 13] Synthesis of R181

(193) ##STR00073##

(194) Under a nitrogen stream, A10 (8.6 g, 18.3 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (9.3 g, 20.1 mmol), Pd(PPh₃)₂Cl₂ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 73.1 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(195) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R181 (10.0 g, 13.7 mmol, yield 75%).

(196) GC-Mass (theoretical value: 729.9 g/mol, measured value: 730 g/mol)

[Synthesis Example 14] Synthesis of R201

(197) ##STR00074##

(198) Under a nitrogen stream, A11 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₂Cl₂ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(199) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R201 (9.0 g, 12.8 mmol, yield 75%).

(200) GC-Mass (theoretical value: 703.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 15] Synthesis of R208

(201) ##STR00075##

(202) Under a nitrogen stream, A11 (7.6 g 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g 18.8 mmol), Pd(PPh₃)₂Cl₂ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(203) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R208 (9.4 g, 12.8 mmol, yield 75%).

(204) GC-Mass (theoretical value: 731.8 g/mol, measured value: 731 g/mol)

[Synthesis Example 16] Synthesis of R221

(205) ##STR00076##

(206) Under a nitrogen stream, A12 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh₃)₂Cl₂ (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(207) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R221 (8.9 g, 12.6 mmol, yield 74%).

(208) GC-Mass (theoretical value: 703.9 g/mol, measured value: 704 g/mol)

[Synthesis Example 17] Synthesis of R241

(209) ##STR00077##

(210) Under a nitrogen stream, A13 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(211) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R241 (9.0 g, 12.8 mmol, yield 75%).

(212) GC-Mass (theoretical value: 702.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 18] Synthesis of R261

(213) ##STR00078##

(214) Under a nitrogen stream, A14 (7.6 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(215) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R261 (8.9 g, 12.0 mmol, yield 74%).

(216) GC-Mass (theoretical value: 702.9 g/mol, measured value: 703 g/mol)

[Synthesis Example 19] Synthesis of R281

(217) ##STR00079##

(218) Under a nitrogen stream, A15 (5.5 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(219) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R281 (9.9 g, 12.8 mmol, yield 75%).

(220) GC-Mass (theoretical value: 577.7 g/mol, measured value: 578 g/mol)

[Synthesis Example 20] Synthesis of R285

(221) ##STR00080##

(222) Under a nitrogen stream, A15 (5.5 g 17.1 mmol), 2-bromo-4,6-bis(dibenzo[b,d]furan-4-yl)-1,3,5-triazine (9.2 g 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(223) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO.sub.4. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R285 (8.0 g, 13.1 mmol, yield 77%).

(224) GC-Mass (theoretical value: 605.7 g/mol, measured value: 606 g/mol)

[Synthesis Example 21] Synthesis of R286

(225) ##STR00081##

(226) Under a nitrogen stream, A16 (5.9 g, 17.1 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (8.7 g, 18.8 mmol), Pd(PPh.sub.3).sub.4 (1.0 g, 5 mol %), potassium carbonate (7.1 g, 51.2 mmol) and toluene/H.sub.2O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(227) After the reaction was finished, the organic layer was separated using methylene chloride,

and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R286 (7.2 g, 11.9 mmol, yield 70%).

(228) GC-Mass (theoretical value: 603.8 g/mol, measured value: 604 g/mol)

[Synthesis Example 22] Synthesis of R291

(229) ##STR00082##

(230) Under a nitrogen stream, A17 (6.1 g, 18.3 mmol), 2-(3'-bromobiphenyl-3-yl)-4,6-diphenyl-1,3,5-triazine (9.3 g, 20.1 mmol), Pd(PPh₃)₄ (1.0 g, 5 mol %), potassium carbonate (7.6 g, 73.1 mmol) and toluene/H₂O/ethanol (80 ml/40 ml/40 ml) were introduced, and stirred for 3 hours at 110° C.

(231) After the reaction was finished, the organic layer was separated using methylene chloride, and water was removed using MgSO₄. After the solvent of the organic layer was removed, the result was purified using column chromatography to obtain target Compound R291 (9.2 g, 13.7 mmol, yield 75%).

(232) GC-Mass (theoretical value: 669.9 g/mol, measured value: 670 g/mol)

Examples 1 to 6 Manufacture of Green Organic Electroluminescent Device

(233) After high purity sublimation purifying the compounds synthesized in the synthesis examples using commonly known methods, green organic electroluminescent devices were manufactured using the following procedure.

(234) First, a glass substrate on which indium tin oxide (ITO) was coated as a thin film to a thickness of 1500 Å was ultrasonic cleaned using distilled water. After the cleaning with distilled water was finished, the substrate was ultrasonic cleaned with solvents of isopropyl alcohol, acetone, methanol and the like, dried, then transferred to a UV OZONE washer (Power sonic 405, manufactured by Hwashin Tech. Co., Ltd.), and then, after cleaning the substrate for 5 minutes using UV, the substrate was transferred to a vacuum deposition apparatus.

(235) On the transparent ITO electrode prepared as above, m-MTDATA (60 nm)/TCTA (80 nm)/each compound of R1, R8, R141, R148, R201 and R208+10% Ir(ppy)₃ (300 nm)/BCP (10 nm)/Alq₃ (30 nm)/LiF (1 nm)/Al (200 nm) were laminated in this order to manufacture an organic electroluminescent device.

(236) Structures of m-MTDATA, TCTA, Ir(ppy)₃, CBP and BCP are as follows.

(237) ##STR00083##

Comparative Example 1 Manufacture of Green Organic Electroluminescent Device

(238) A green organic electroluminescent device was manufactured in the same manner as in Example 1 except that CBP was used instead of Compound R141 as the light emitting host material when forming the light emitting layer.

Evaluation Example 1

(239) For each of the green organic electroluminescent devices manufactured in Examples 1 to 6 and Comparative Example 2, driving voltage, current efficiency and light emission peak at current density of (10) mA/cm² were measured, and the results are shown in the following Table 1.

(240) TABLE-US-00001 TABLE 1 Driving Current Voltage EL Peak Efficiency Sample Host (V) (nm) (cd/A) Example 1 R1 5.2 515 12.5 Example 2 R8 4.3 515 11.4 Example 3 R141 5.1 515 12.3 Example 4 R148 4.2 515 11.1 Example 5 R201 5.4 515 12.2 Example 6 R208 4.4 515 10.8 Comparative CBP 7.1 516 7.2 Example 1

(241) As shown in Table 1, it was seen that using the compound (R1, R8, R141, R148, R201, R208) according to the present invention in a light emitting layer of the green organic EL device (Examples 1 to 6) resulted in more superior performance in terms of efficiency and driving voltage compared to the green organic EL device using existing CBP (Comparative Example 1).

Examples 7 to 28 Manufacture of Blue Organic Electroluminescent Device

(242) After high purity sublimation purifying the compounds synthesized in the synthesis examples using commonly known methods, blue organic electroluminescent devices were manufactured

using the following procedure.

(243) First, a glass substrate on which indium tin oxide (ITO) was coated as a thin film to a thickness of 1500 Å was ultrasonic cleaned using distilled water. After the cleaning with distilled water was finished, the substrate was ultrasonic cleaned with solvents of isopropyl alcohol, acetone, methanol and the like, dried, then transferred to a UV OZONE washer (Power sonic 405, manufactured by Hwashin Tech. Co., Ltd), and then, after cleaning the substrate for 5 minutes using UV, the substrate was transferred to a vacuum deposition apparatus.

(244) On the transparent ITO electrode prepared as above, DS-205 (Doosan Corporation Electro-Materials, 80 nm)/NPB (15 nm)/ADN+5% DS-405 (Doosan Corporation Electro-Materials, 30 nm)/each compound of R1, R8, R21, R41, R61, R68, R81, R101, R121, R141, R141, R148, R161, R181, R201, R208, R221, R241, R261, R281, R285, R286 and R291 (5 nm)/Alq.sub.3 (25 nm)/LiF (1 nm)/Al (200 nm) were laminated in this order to manufacture an organic electroluminescent device.

Comparative Example 2 Manufacture of Blue Organic Electroluminescent Device

(245) A blue organic electroluminescent device was manufactured in the same manner as in Example 6 except that R1 was not used as the electron transport auxiliary layer material, and Alq.sub.3, the electron transport layer material, was deposited to 30 nm instead of 25 nm.

(246) Structures of NPB, AND and Alq.sub.3 used in Examples 6 to 28 and Comparative Example 2 are as follows.

(247) ##STR00084##

Comparative Example 3 Manufacture of Blue Organic Electroluminescent Device

(248) A blue organic electroluminescent device was manufactured in the same manner as in Example 6 except that A1 of the following structural formula was used instead of R1 as the electron transport auxiliary layer material.

(249) ##STR00085##

Evaluation Example 2

(250) For each of the organic electroluminescent devices manufactured in Examples 7 to 28 and Comparative Examples 2 and 3, driving voltage, light emission wavelength and current efficiency at current density of 10 mA/cm² were measured, and the results are shown in the following Table 2.

(251) TABLE-US-00002

TABLE 2	Electron	Driving	Light	Current	Transport	Voltage	Emission				
Efficiency	Sample	Auxiliary	Layer	(V)	Peak	(nm)	(cd/A)				
Example 7	R1	4.4	458	8.3	Example 8	R8	3.9				
458	7.5	Example 9	R21	4.4	458	8.0	Example 10	R41	4.4	458	8.2
Example 11	R61	4.4	458	8.1	Example 12	R68	4.4	458	8.2		
Example 13	R81	4.4	458	8.3	Example 14	R101	4.4	458	8.0		
Example 15	R121	4.4	458	8.2	Example 16	R141	4.3	458	7.8		
Example 17	R148	3.5	458	7.1	Example 18	R161	4.4	458	7.3		
Example 19	R181	4.5	458	7.4	Example 20	R201	4.1	458	7.5		
Example 21	R208	3.9	458	7.0	Example 22	R221	4.2	458	7.5		
Example 23	R241	4.1	458	7.6	Example 24	R261	4.5	458	7.3		
Example 25	R281	3.8	458	6.9	Example 26	R285	3.4	458	6.8		
Example 27	R286	4.7	458	7.2	Example 28	R291	3.8	458	6.9		
Comparative	Alq.sub.3	4.8	458	6.2	Example 2	Comparative	A1	4.7	457	6.5	

Example 3

(252) As shown in Table 2, it was seen that the blue organic electroluminescent devices using the compound of the present invention in an electron transport auxiliary layer (Examples 6 to 28) exhibited superior performance in terms of current efficiency, light emission peak and driving voltage compared to the blue organic electroluminescent devices without an electron transport auxiliary layer (Comparative Examples 2 and 3).

Examples 29 to 32 Manufacture of Blue Organic Electroluminescent Device

(253) After high purity sublimation purifying the compounds synthesized in the synthesis examples using commonly known methods, blue organic electroluminescent devices were manufactured using the following procedure.

(254) First, a glass substrate on which indium tin oxide (ITO) was coated as a thin film to a

thickness of 1500 Å was ultrasonic cleaned using distilled water. After the cleaning with distilled water was finished, the substrate was ultrasonic cleaned with solvents of isopropyl alcohol, acetone, methanol and the like, dried, then transferred to a UV OZONE washer (Power sonic 405, manufactured by Hwashin Tech. Co., Ltd.), and then, after cleaning the substrate for 5 minutes using UV, the substrate was transferred to a vacuum deposition apparatus.

(255) On the transparent ITO electrode prepared as above, DS-205 (Doosan Corporation Electro-Materials, 80 nm)/NPB (15 nm)/ADN+5% DS-405 (Doosan Corporation Electro-Materials, 30 nm)/each compound of R1, R8, R21 and R41 (30 nm)/LiF (1 nm)/Al (200 nm) were laminated in this order to manufacture an organic electroluminescent device.

Comparative Example 4 Manufacture of Blue Organic Electroluminescent Device

(256) A blue organic electroluminescent device was manufactured in the same manner as in Example 29 except that Alq.sub.3 was used instead of R1 as the electron transport layer material.

Comparative Example 5 Manufacture of Blue Organic Electroluminescent Device

(257) A blue organic electroluminescent device was manufactured in the same manner as in Example 29 except that R1 was not used as the electron transport layer material.

Evaluation Example 3

(258) For each of the blue organic electroluminescent devices manufactured in Examples 29 to 32 and Comparative Examples 2 and 3, driving voltage, current efficiency and light emission wavelength at current density of 10 mA/cm² were measured, and the results are shown in the following Table 3.

(259) TABLE-US-00003 TABLE 3 Electron Driving Light Current Transport Voltage Emission Efficiency Sample Layer (V) Peak (nm) (cd/A) Example 29 R1 4.5 455 8.7 Example 30 R8 3.9 455 7.8 Example 31 R21 4.4 455 8.4 Example 32 R41 4.2 455 9.3 Comparative Alq.sub.3 4.7 458 5.5 Example 4 Comparative — 4.8 460 6.2 Example 5

(260) As shown in Table 3, it was seen that the blue organic electroluminescent devices using the compound of the present invention in an electron transport layer (Examples 29 to 32) exhibited superior performance in terms of driving voltage, light emission peak and current efficiency compared to the blue organic electroluminescent device using existing Alq.sub.3 in an electron transport layer (Comparative Example 4), and the blue organic electroluminescent device without an electron transport layer (Comparative Example 5).

REFERENCE NUMERAL

(261) **10**: Anode **20**: Cathode **30**: Organic Layer **31**: Hole Transport Layer **32**: Light Emitting Layer **33**: Hole Transport Auxiliary Layer **34**: Electron Transport Layer **35**: Electron Transport Auxiliary Layer **36**: Electron Injection Layer **37**: Hole Injection Layer

Claims

1. A compound selected from the group consisting of the following compounds: ##STR00086## ##STR00087## ##STR00088## ##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093## ##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098## ##STR00099## ##STR00100## ##STR00101## ##STR00102##
 2. An organic electroluminescent device comprising: an anode; a cathode; and one or more organic material layers provided between the anode and the cathode, wherein at least one of the one or more organic material layers includes a compound of claim 1.
 3. The organic electroluminescent device of claim 2, wherein the organic material layer includes one or more layers selected from the group consisting of a hole injection layer, a hole transport layer, a hole transport auxiliary layer, an electron transport layer, an electron transport auxiliary layer and a light emitting layer.
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